VIJAYALAKSHMI

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Profile Summary

Dynamic and results-oriented **Data Engineer** with 4+ years of experience delivering scalable **cloud-based data pipelines**, real-time dashboards, and automation solutions across **Retail E-commerce** and **Pharma** domains. Skilled in **Python**, **SQL (Redshift, Snowflake)**, and **AWS (Glue, Lambda, S3)**, with a strong focus on **ETL development**, **data modelling**, and **BI reporting** using **Power BI**. Proven ability to optimize workflows, enhance data quality, and enable faster, smarter decision-making. Passionate about leveraging automation and cloud technologies to drive **insightful, innovation-led outcomes**.

Technical Skills

Programming Languages	: Python, PySpark, SQL
Data Manipulation & Analysis	: Pandas & NumPy
ETL / ELT Tools	: AWS (S3, Glue, Redshift, IAM), Databricks, Python ETL
Cloud Platforms	: AWS, Azure
Data Visualization	: Power BI, Tableau, Matplotlib
Data Wrangling	: Data Cleaning, Transformation, Feature Engineering
Data Warehousing	: MySQL, Snowflake
Version Control / DevOps	: Git, AWS DevOps
Database Management	: AWS RDS, MySQL, Redshift SQL
Excel	: Advanced Functions, PivotTables, Power Query
Data Governance & Security	: IAM Policies, RLS, HIPAA/GDPR Compliance, Data Masking, Audit Logging

Professional Experience

TGK Streamline Solutions, Chennai, India

Data Engineer | Aug 2021 – Present (**4 Years**)

- Participate in daily stand-ups to update progress on data pipeline development, report blockers, and align with engineering tasks.
- Attend sprint planning and retrospectives to estimate ETL workload, review performance of data jobs, and suggest process improvements.
- Prepare documentation and walkthroughs for ETL workflows, data models, and architecture to enable smooth knowledge transfer.

- Ensure data pipelines and storage follow company policies, governance standards, and compliance frameworks (HIPAA/GDPR).
- Collaborate with cross-functional teams (data analysts, BI developers, data scientists) to deliver reliable, production-ready datasets.

Projects

Pharma Data Platform – AWS Cloud & BI Stack Project

Duration: Aug 2023 | Present

Summary:

Architected and implemented a secure, compliant data ecosystem for pharmacy prescription tracking and fulfillment analytics leveraging AWS, Snowflake, and Power BI.

Key Responsibilities & Achievements:

- Designed and managed end-to-end data pipelines using AWS Glue, Lambda, Step Functions, and Redshift to automate ingestion of pharmacy data from S3 and APIs.
- Modelled data marts with star and snowflake schemas, supporting flexible BI reporting for pharmacy performance, customer retention, and medication analytics.
- Built interactive Power BI dashboards with custom DAX metrics, RLS, and automated refresh, improving visibility into SLAs, out-of-stock alerts, and regional performance.
- Used Redshift Spectrum and S3 partitioning to optimize cost-effective querying and storage.
- Developed AWS Glue jobs in PySpark to clean and transform prescription data from S3 to Snowflake and Redshift.
- Used AWS Glue Crawlers to automatically infer schema and catalog new datasets.
- Implemented monitoring and alerting with SNS, CloudWatch, and retry logic in ETL jobs, reducing pipeline downtime by 80%.
- Designed reusable ETL modules using Lambda layers, improving maintainability and reducing onboarding time for new use cases.
- Developed and optimized complex SQL queries in Redshift and Snowflake for medication trends, prescription duplication checks, and anomaly detection using window functions and CTEs.
- Created automated audit trails using CloudTrail, S3 logs, and metadata tagging in the Glue Data Catalog.

- Delivered a pharmaceutical data platform that improved reporting efficiency by 60%, ensured regulatory compliance (HIPAA, PII), and reduced manual data operations by over 70%
- Used DAX to create custom measures for revenue, inventory level, and fulfilment rates.

Retail E-commerce Analytics & Data Engineering Project

Duration: Aug 2021 – July 2023

Summary:

Built an automated data platform using AWS, Python, and BI tools to streamline e-commerce analytics, enabling deeper insights into sales, customer behavior, and inventory management.

Key Responsibilities & Achievements:

- Engineered and automated daily ETL pipelines to ingest data using Python, AWS Glue, and Redshift, improving reporting latency by 70%.
- Developed custom data quality checks using Cerberus and regex, reducing bad data ingestion by 40% and increasing trust in analytics.
- Created dynamic dashboards to track sales trends and inventory metrics.
- Created interactive dashboards in Power BI and Tableau to track KPIs such as Average Order Value (AOV), Customer Lifetime Value (CLV), and conversion rates by region and category.
- Built scalable PySpark Glue jobs for processing inventory and order data, with downstream loading to Redshift Spectrum for efficient querying.
- Automated multi-region data handling with parameterized Python CLI tools and argparse, supporting region-specific pipelines and reducing manual configuration.
- Modelled analytics layers in star schema, enabling self-service analytics for business teams and improving dashboard load performance.
- Deployed event-driven pipelines using AWS Lambda, Step Functions, and Event Bridge to manage near-real-time data processing.
- Applied secure access and version control through Lake Formation, IAM, and S3 lifecycle policies, maintaining compliance and optimizing storage costs.
- Performed time-series and cohort analysis using Pandas, supporting marketing and retention strategy refinements.

- Automated Excel-based reports for operations teams using OpenPyXL, reducing manual reporting hours by over 60%.

Research Project

Telecom Fraud Detection Using LSTM with NADAM Optimizer
Tools Used: Python, TensorFlow, Autoencoder, LSTM, NADAM, K-Means, SelectKBest

- Built a deep learning model using LSTM with NADAM optimizer to detect fraudulent telecom activities such as robocalls and spoofing from CDR data.
- Applied Autoencoder for feature reduction and K-Means + SelectKBest for optimal feature selection, improving model efficiency.
- Achieved 98% accuracy, 97% recall, and 91% precision on 850K+ call records through optimized sequence modeling and preprocessing.
- Enabled real-time fraud classification and spam call prediction for financial sector telecom data using scalable deep learning techniques.

Education

B.Sc - Bachelor of Science in Computer Science
Vel Sri Ranga Sanku College, Madras University - Gold Medalist | 2011

MCA - Master of Computer Applications
Meenakshi College of Engineering, Anna University | 2015

Ph.D. in Computer Science (Pursuing since 2024)
Vels Institute of Science, Technology & Advanced Studies